

Reproducing SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning

Introduction:

Few-shot learning aims to classify novel visual categories from only a few labeled examples, and many prior methods rely on complex meta learning procedures to avoid overfitting when adapting to unseen classes. In *SimpleShot: Revisiting Nearest Neighbor Classification for Few Shot Learning*, Yan Wang, Wei Lun Chao, Kilian Q. Weinberger, and Laurens van der Maaten [1] challenge this assumption by showing that a simple nearest neighbor or nearest centroid classifier can be highly competitive when applied to strong CNN features. Instead of introducing a complex meta learning algorithm, the paper emphasizes the importance of feature space geometry: standard cross entropy trained CNN features can support strong few shot classification if they are transformed with L2 normalization or centered L2 normalization before computing Euclidean distances. In this project, we reproduce the paper's central claim on minilImageNet and examine whether its main qualitative conclusions remain robust under our own implementation and modern training environment.

Chosen Result:

We chose to reproduce the main minilImageNet result from Table 1 of *SimpleShot* [1], which reports five way one shot and five shot classification accuracy across multiple backbone networks and feature transformations. Specifically, we evaluate Conv4, ResNet10, ResNet 18, and DenseNet 121 with three feature representations: unnormalized features (UN), L2 normalized features (L2N), and centered L2 normalized features (CL2N). This result was selected because it directly tests the paper's main empirical claim that properly normalized CNN features can make simple nearest neighbor or nearest centroid classifiers strong few shot learners. It also allows us to study several reproduction questions in one setting: whether SimpleShot works reliably on minilImageNet, whether deeper backbones consistently improve performance, how much L2N and CL2N improve over UN, how best checkpoint results differ from final epoch results, and what factors may explain any remaining accuracy gap between our reproduced results and the original paper.

Methodology:

We examined the original GitHub implementation, the SimpleShot paper [1], and related works on few shot evaluation and backbone architectures [2,3,4,5] to reconstruct the training and evaluation pipeline. We reimplemented SimpleShot on minilImageNet with four backbones: Conv 4, ResNet 10, ResNet 18, and DenseNet 121. Following the original setup, each model was first trained on the 64 base classes as a standard supervised classifier with cross entropy loss. We trained each backbone for 90 epochs using SGD with momentum 0.9, weight decay 1×10^{-4} , batch size 256, and LR drops at epochs 45 and 66. Training used common image augmentations, including random resized crop, horizontal flip, and ImageNet normalization. After training, the final FC classifier was removed, and the backbone was used as a fixed feature extractor for few shot evaluation.

For evaluation, we sampled 10,000 five way episodes from the 20 novel minilImageNet test classes. Each episode used either 1 shot or 5 shot support examples per class and 15 query images per class. In the 1 shot setting, each query was classified by Euclidean nearest neighbor over support features. In the 5 shot setting, we averaged support features within each class to form class centroids and assigned each query to the nearest centroid. Before distance computation, we tested three feature transformations: UN, L2N, CL2N, where CL2N subtracts the base class feature mean before L2 normalization. To study checkpoint effects, we reported both final epoch and best checkpoint results.

Results & Analysis:

Tables 1 and 2 compare our reproduced minImageNet results against the original SimpleShot paper using both the best validation checkpoint and the final epoch 90 checkpoint. Overall, our implementation successfully reproduces the paper’s main qualitative findings. Across both 1-shot and 5-shot evaluations, deeper backbones consistently achieve higher accuracy than shallower architectures. Conv4 produces the weakest performance, while DenseNet121 achieves the strongest results, indicating that stronger supervised feature representations directly improve few-shot generalization.

Feature normalization also has a significant impact on performance. Across nearly all backbone architectures, L2-normalized (L2N) features outperform unnormalized (UN) features, while centered L2-normalized (CL2N) features usually provide the best accuracy overall. This supports the paper’s central claim that the geometry of the feature space is more important than complex meta-learning procedures.

Although the overall trends match the original paper, our reproduced accuracies are generally 2–3% lower than the reported values for stronger backbones. Several factors likely contribute to this gap. First, the original implementation was developed under an older PyTorch environment (PyTorch 1.0+), while our experiments used a more modern training stack (PyTorch 2.0+). Second, subtle differences in image preprocessing, random episode sampling, and augmentation pipelines can noticeably affect few-shot benchmarks.

Another important observation is that the performance difference between the best checkpoint and the final epoch 90 checkpoint is relatively small across all settings. This suggests that the training process is stable and that the models do not significantly overfit late in training. As a result, SimpleShot appears less sensitive to checkpoint selection than many meta-learning-based approaches.

Reflections:

This project provided valuable insight into both few-shot learning and the practical challenges of reproducing research. One of our main takeaways is that relatively simple methods can remain highly competitive when paired with strong feature representations and carefully designed preprocessing. Before reproducing SimpleShot, we expected that complex meta-learning algorithms would be necessary for strong few-shot performance, but our experiments demonstrated that feature quality and feature-space geometry alone can achieve surprisingly strong results.

We also learned that reproducing published results is often difficult, even when the overall methodology appears straightforward. Small implementation details, including preprocessing choices, software versions, and training schedules, can all produce measurable differences in final accuracy. This experience highlighted the importance of reproducibility and detailed experimental reporting in deep learning research.

If given more time and computational resources, we would extend this work in several directions. First, we would evaluate SimpleShot on a few additional few-shot benchmarks such as CIFAR-FS, tieredImageNet, and CUB to test whether the same trends generalize across datasets. Second, we would explore modern backbone architectures such as Vision Transformers and ConvNeXt models within the SimpleShot framework. Finally, we would further investigate the geometric effect of CL2N normalization on class separation and embedding distributions to better understand why such a simple modification improves few-shot classification performance.

Appendix I: Experimental Results

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.11+/-0.17	33.17+/-0.17	+0.94	62.19+/-0.17	63.25+/-0.17	-1.06
conv4	L2N	46.34+/-0.18	48.08+/-0.18	-1.74	64.52+/-0.17	66.49+/-0.17	-1.97
conv4	CL2N	47.98+/-0.19	49.69+/-0.19	-1.71	65.12+/-0.17	66.92+/-0.17	-1.80
resnet10	UN	52.01+/-0.21	54.45+/-0.21	-2.44	74.39+/-0.16	76.98+/-0.15	-2.59
resnet10	L2N	55.76+/-0.20	57.85+/-0.20	-2.09	75.77+/-0.15	78.73+/-0.15	-2.96
resnet10	CL2N	58.33+/-0.20	60.85+/-0.20	-2.52	75.75+/-0.16	78.40+/-0.15	-2.65
resnet18	UN	53.90+/-0.21	56.06+/-0.20	-2.16	76.47+/-0.15	78.63+/-0.15	-2.16
resnet18	L2N	58.05+/-0.20	60.16+/-0.20	-2.11	77.59+/-0.15	79.94+/-0.14	-2.35
resnet18	CL2N	60.34+/-0.20	62.85+/-0.20	-2.51	77.61+/-0.15	80.02+/-0.14	-2.41
densenet121	UN	55.40+/-0.20	57.81+/-0.21	-2.41	78.17+/-0.15	80.43+/-0.15	-2.26
densenet121	L2N	59.64+/-0.20	61.49+/-0.20	-1.85	78.81+/-0.14	81.48+/-0.14	-2.67
densenet121	CL2N	61.94+/-0.20	64.29+/-0.20	-2.35	79.08+/-0.15	81.50+/-0.14	-2.42

Note: Negative gap = below paper. Differences due to PyTorch version + pickle data source.
 Paper used PyTorch 1.0 + original jpg images. We use PyTorch 2.x + Kaggle pickle.
 See GitHub Issue #5: PyTorch 1.4 already causes 2-3% drop vs PyTorch 1.0.

Table 1: Best checkpoint minilImageNet reproduction results compared with the original SimpleShot Table 1.

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.28+/-0.17	33.17+/-0.17	+1.11	62.35+/-0.17	63.25+/-0.17	-0.90
conv4	L2N	46.50+/-0.18	48.08+/-0.18	-1.58	64.48+/-0.17	66.49+/-0.17	-2.01
conv4	CL2N	48.03+/-0.19	49.69+/-0.19	-1.66	65.20+/-0.17	66.92+/-0.17	-1.72
resnet10	UN	51.86+/-0.21	54.45+/-0.21	-2.59	74.46+/-0.16	76.98+/-0.15	-2.52
resnet10	L2N	55.68+/-0.20	57.85+/-0.20	-2.17	75.81+/-0.16	78.73+/-0.15	-2.92
resnet10	CL2N	58.39+/-0.20	60.85+/-0.20	-2.46	75.80+/-0.16	78.40+/-0.15	-2.60
resnet18	UN	53.82+/-0.21	56.06+/-0.20	-2.24	76.56+/-0.15	78.63+/-0.15	-2.07
resnet18	L2N	57.72+/-0.20	60.16+/-0.20	-2.44	77.70+/-0.15	79.94+/-0.14	-2.24
resnet18	CL2N	60.30+/-0.20	62.85+/-0.20	-2.55	77.83+/-0.15	80.02+/-0.14	-2.19
densenet121	UN	55.54+/-0.20	57.81+/-0.21	-2.27	77.95+/-0.15	80.43+/-0.15	-2.48
densenet121	L2N	59.55+/-0.20	61.49+/-0.20	-1.94	78.60+/-0.15	81.48+/-0.14	-2.88
densenet121	CL2N	61.78+/-0.20	64.29+/-0.20	-2.51	79.10+/-0.15	81.50+/-0.14	-2.40

Note: Negative gap = below paper. Differences due to PyTorch version + pickle data source.
 Paper used PyTorch 1.0 + original jpg images. We use PyTorch 2.x + Kaggle pickle.
 See GitHub Issue #5: PyTorch 1.4 already causes 2-3% drop vs PyTorch 1.0.

Table 2: epoch 90 minilImageNet reproduction results compared with the original SimpleShot Table 1.

Appendix II: References

- [1] Y. Wang, W.-L. Chao, K. Q. Weinberger, and L. van der Maaten, "SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning," *arXiv preprint arXiv:1911.04623*, 2019.
- [2] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching Networks for One Shot Learning," in *NeurIPS*, 2016.
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- [4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.
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